

EVSP DR Current Research

12 September 2026

Inherited covering matches $k=3-6$ in all six chains.
Three chains now match ten buses at $k=10$.

Remaining questions: initialization cost, capacity pricing
and whether decomposition helps larger instances.

Fresh starts need extra buses as k grows

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
5	5	5	5	5	6	5
8	9	9	9	9	9	8
10	11	11	11	11	11	11
15	18	17	18	19	16	19

Integer incumbents from all 84 completed fresh covering cases.

Each size starts independently. MIP proof status varies by cell.

The current document contains the full chain tables and source records.

Inherited columns improve integer fleet

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
2	2	2	2	2	3	2
5	5	5	5	5	5	5
6	6	6	6	6	6	6
8	—	8	8	8	8	8
10	—	—	10	—	10	10

Cells show integer buses. Grey means unfinished. k=11–15 is running.
All displayed fleet optima are proved within their saved pools.
240 kWh / 240 kW, covering. Shared charger capacity is not imposed.

Station capacity exposes a pricing bottleneck

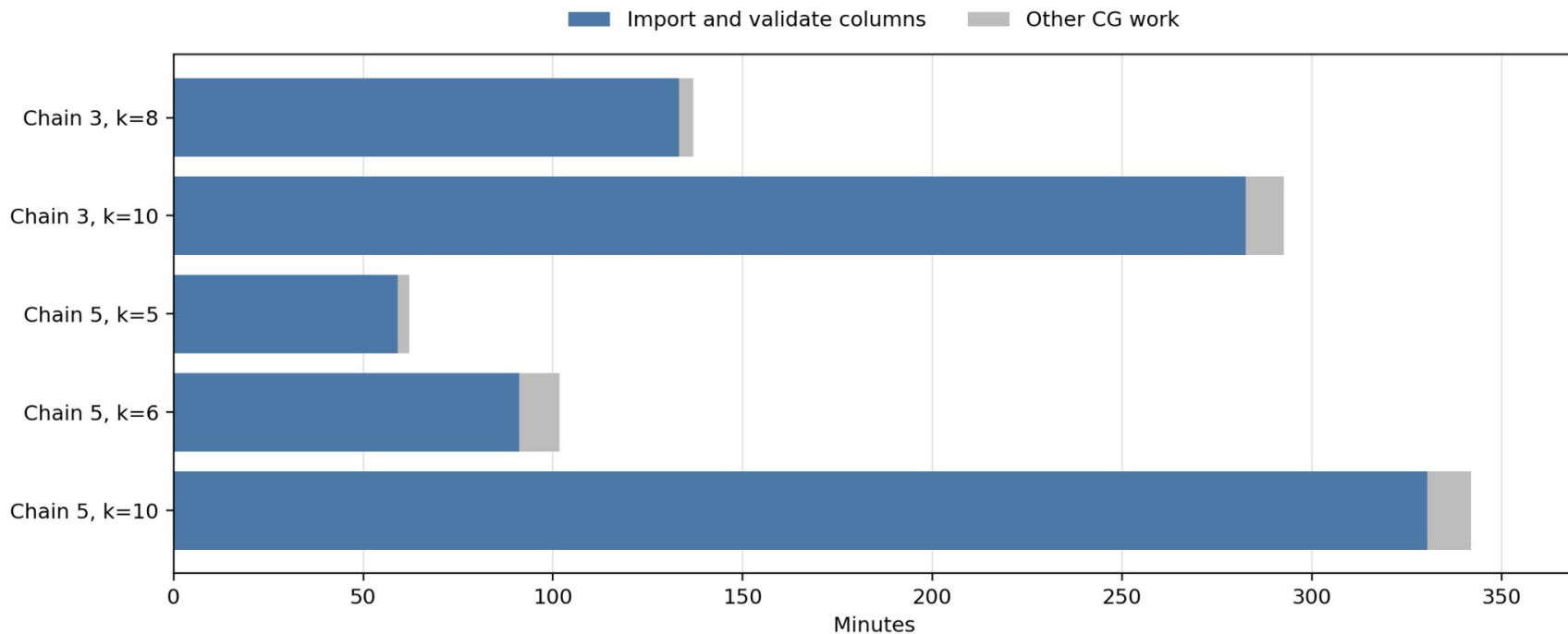
Target	Baseline	PARX 60 kW	Capacity	Both
1, duty A	1	1	1	1
1, duty B	1	1	1	1
2	2	2	3	3
3	3	3	16	16

k=3: one pricing call took 7.15 hours; the LP took 0.006 seconds.

CG stopped uncertified with only three new columns.

The 16-bus pool result does not prove that 16 buses are necessary.

Importing columns dominates warm-run time



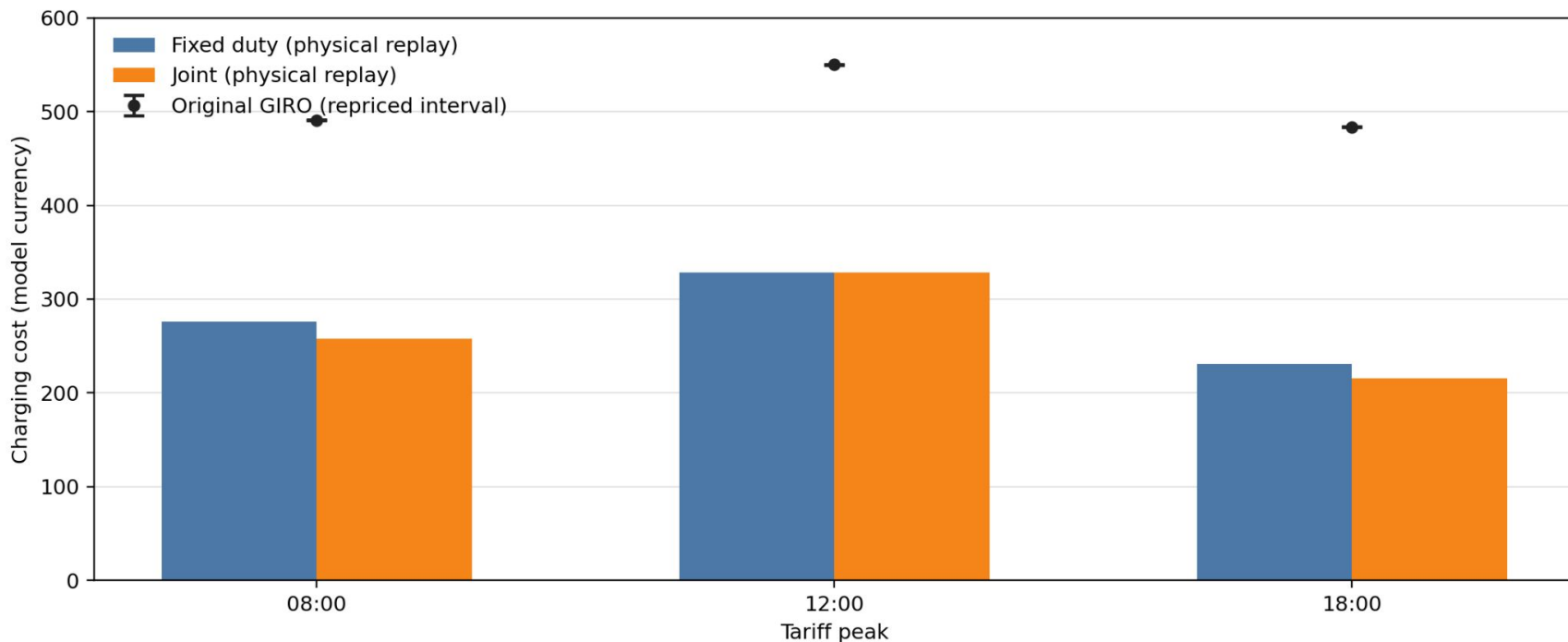
Full predecessor pools were replayed here. The new overnight treatment caps replay at 512 sequences / 15 minutes; results are pending.

The same LP can support different integer pools

Chain 3 at k=8	Fresh pool	Inherited pool
Weighted LP objective	800,212.68747	800,212.68747
Fractional route weight	8	8
Integer fleet optimum in pool	9	8

Both CG runs met the same reduced-cost stopping tolerance.
The unchanged fresh pool cannot reach eight buses with more MIP time.
This isolates a column-pool limitation for this case.

Charging savings with matched return energy



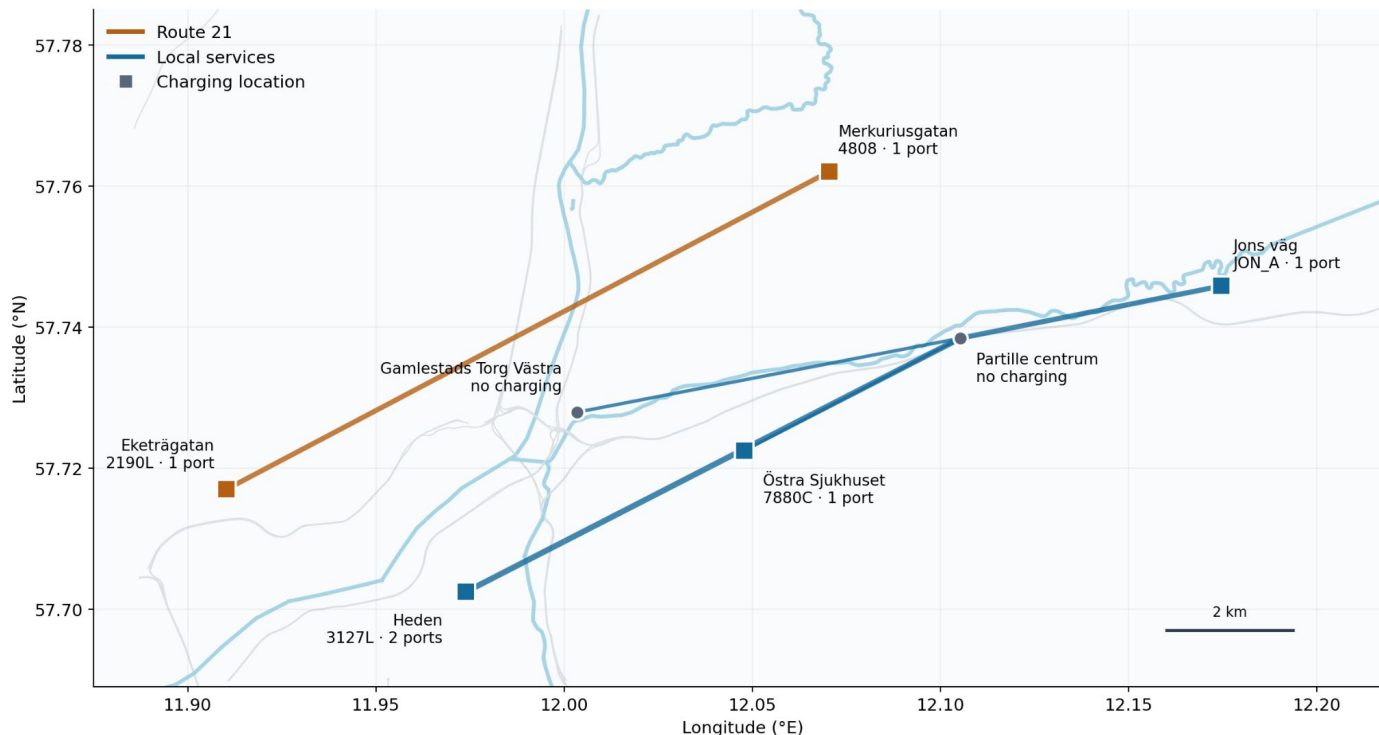
Five buses / 62 trips. Both optimized cases return ≥ 280.7833 kWh in total. At noon, grid objectives tie; physical replay costs differ slightly.

The overnight experiments answer three questions

Question	Experiment
Can inheritance scale?	Six chains continue to $k=15$
Does decomposition help?	Ten partitions of 32 duties into groups of eight
Can recombination improve it?	Global CG starts from component routes

87 CG cases and 87 dependent one-hour MIPs, all on default_partition.
Bounded replay is a new treatment: ≤ 512 sequences and ≤ 15 minutes.
No new $k=11-15$ or decomposition result is claimed yet.

Geography suggests groups, not independence



838 of 948 trips mapped; lines join endpoints, not road paths. Timetable connections still cross groups. Map data: OpenStreetMap.